

# AUTO\_RESEARCH.md — the workshop as a public-goods research instrument

*The frame the First Person named: this build is, in theory, a public-goods donation to a university in an auto-research pattern. This doc says what that means mechanically, which patterns from the wider work are woven in, and which are staged next. It is the bridge between the course (CURRICULUM.md) and the letters in `Letters/`.*

## 1 • The donation, precisely

What the university receives is not a syllabus. It is a **self-improving research instrument** with its provenance attached:

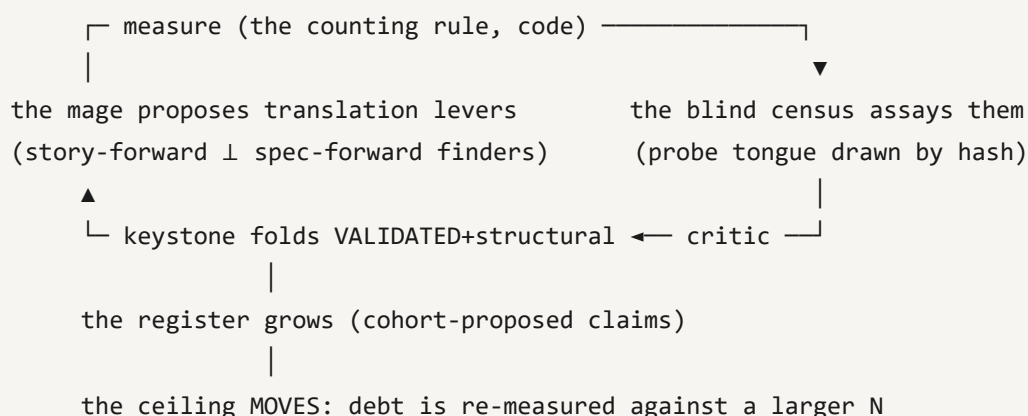
- **the instrument** — this repository: a verification harness whose gate is blind and replayable, a local mage whose training corpus is consent-gated at the pipe, a law lane that compiles, and a control surface (the Towel) that can only do what the repo's own scripts do;
- **the method** — the babblefish recursion as a research protocol: understanding is only counted when it survives translation across tongues, checked by an assayer who never sees the canon;
- **the record** — everything the instrument produces is a citable artifact: the amended claim register, validated triptychs with offline-replayable verdicts, the steward chain, the bound books, each cohort's mage.

Public-goods properties, by construction: zero marginal dependency (no cloud, no subscription, no API key — the machine in the flight case is the whole stack); **non-extractive** (the absence claims in `lexon/` make extraction *structurally impossible*, not merely forbidden); forkable (the engine's upstream is public; a sister department

copies the directory and owns its own instrument); and **the students are not the product — they are the provenance** (artifact/PROVENANCE.md).

## 2 • The auto-research pattern

The loop research runs on its own; humans hold the folds and the doors.



Three patterns from the wider work are load-bearing here:

- **The moving ceiling (V6's R(t))** — research never "completes." When the cohort folds new claims into the register, 3·N grows, debt jumps, and the frontier advances against a *raised* target. An honest instrument reports the ceiling's motion, not just the score. `frontier.json` is the sole authority; OT-numbered open targets carry the "next number to beat."
- **The dream cycle (measure-only upkeep)** — between sessions the instrument can run unattended in the safest posture: `measure + verify_run --all + corpus_pack --verify + lexon/check` on a schedule. A standing watch that surfaces gaps and drift, folds nothing, opens no door. (The `lexon_pvm runtime-01` "standing watch" and the fleet's dream cycle are the precedents.) Proposal rounds can also run unattended — verdicts queue as *candidates*; folding stays a human act at the next session. That split — **automate the measuring and the proposing, never the folding or the door** — is the whole pattern.

- **Chronicles as the research record** — every session ends in a chronicle (GR-7). For a university this is the lab notebook, pre-formatted: verdict-first, reversals at win-prominence, a handoff naming the next lead.

### 3 • Weaving the fine-tuning inference system

The mage is one Ollama endpoint wearing three hats, and the weave is that the hats share provenance:

1. **Facilitator** (inference) — `hh-mage`, the base model grounded in the consent-gated context pack; the mirror the room talks to.
2. **The seats** (inference-as-verification) — the same endpoint holds `measure/propose/gap/assay/critic/chronicle` in the harness rounds.  
 $\Phi_{\text{inference}} = 0$  on a one-box room and is *declared*, never hidden; a second box (or a visiting model) raises it — a pre-registered experiment any cohort can run.
3. **The trained lineage** (fine-tuning) — each steward fold advances the mage's grounding; each cohort's endpoint state is reproducible from `steward_log.md` + the corpus at that hash. When option B (LoRA) lands, each cohort mints a **generation**: `hh-mage-c1`, `hh-mage-c2` ... — and the lineage itself becomes the research object: - *ablation across generations* — does cohort-2's mage assay cohort-1's

trptychs differently? (a measurable claim about what the room's words did to the weights); - *cross-cohort  $\Phi$*  — cohort-N's proposer assayed by cohort-M's prover is a genuinely distinct-prior pair: the separation term the PVM wants, grown organically; - *the erasmus weave* — sister universities exchange **methods and probes, never corpora**: a visiting mage assays the local triptychs (blind, offline-verifiable) and goes home. The corpus never travels; the verdicts do. That is the "global erasmus" with sovereignty intact.

## 4 • The Game of 42 lane — persona → geometry → trust graph

The chain the First Person named, now wired ( `tools/workshop_ui.mjs` , `/game/` ):

1. **Persona + skill matching** — students join the roster (consent-gated: no ledger row, no seat) declaring a persona and their tongues. Pair matching is deterministic and hash-drawn — the Gap's idiom applied socially: nobody tunes who they're paired with, and complementary tongues pair first (the finder pair, in human form).
2. **Geometry** — the vendored Game of 42 board (6 axes × 7 stations; the City of Mages seed preset "meets you where you're up to"). Each node is a trust task: assign an agent or yourself. The cohort's weekly seal is the fold — the room's state made one star-form,  $\kappa$ -addressed, dated.
3. **The trust graph** — earned, never declared: workshop events mint ed25519 `did:key` edges over  $\kappa$ -addressed persona holons ( `tools/vrc.mjs` ) — mentor-assay, steward-fold, pair-swap, keystone-fold, seal-witness. No self-edges (C30: a trust that certifies itself). Every edge verifies offline. By course end the cohort holds a small, real, replayable trust graph *that grew out of verified work* — the  VRC pattern at seminar scale, and the on-ramp to the wider graph (hitchhikers.earth's Proof of Understanding; the agentprivacy trust communities) — federated later, at the door, by the bearer.

The sci-fi carries all of it: the canon claims are taught through "The Name That Died" before they are ever stated formally; the game's personas give the room its cast; the serious iterations (PVM, IEEE 7012, Lexon, the harness trusts T1–T6) stand behind every mechanic the students touch. Story first, spec always underneath — that ordering is the babblefish thesis.

## 5 • Staged next (named, not built)

- **scheduled dream cycle** — a cron/loop invoking the measure-only watch; trivial once the mage box has a clock discipline (the scripts exist; the schedule is

a venue decision).

- **cohort LoRA generations** (MACHINE\_SETUP.md option B) — hardware-gated.
- **cross-university probe exchange** — needs one sister instance to exist.
- **seal → City-Key bridge** — the game's sealed PNG already carries a cityKey projection; binding cohort seals into the wider identity surface is the bearer's outward act.
- **Lexon as the fourth tongue** — advanced cohorts render a triptych claim as a compiling law; the checker lane is already green and waiting.